

Multi-Source Pandemic Data Visualization and Synchronization for Information Extraction

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Abstract—Visualization is integral to investigating information hidden in data and providing users with intuitive feedback for decision-making. No matter the field a data set describes, inspecting the data visually will yield fruitful insights into the trends and statistics. Over the past calendar year, COVID-19 vaccines have become increasingly available for much of the population. However, the CDC (Centers for Disease Control and Prevention) fails to consider multiple sets of pandemic data in a side-by-side view and synchronize multiple key factors in one web page, limiting medical professionals and individuals to seeing, comparing, and interacting with complete data visualization. To analyze the coronavirus and vaccination data collected from multiple sources, effectively displaying them is critically important for interpreting the pandemic transmission pattern and vaccine efficiency. This paper presents new algorithms for innovative data visualizations that provide users with intuitive feedback and enable them to see a complete story of where the data is concerned. The information derived from our developed web-based data visualization will aid healthcare professionals and everyday citizens in moving forward as the pandemic progresses.

Index Terms—Synchronization, Data-Driven Documents (D3), dynamic visualization, data analysis, information extraction, feature enhancement

I. INTRODUCTION

In the spring of 2019, a novel coronavirus that can cause a highly contagious disease called coronavirus disease 2019 (COVID-19), challenged the world on scales many could not have foreseen, affecting almost all countries and territories. Due to the uncertainty that the world is facing, dynamic, reliable, and accurate information is needed now more than ever. The virus has gone from a disease located in a remote region of the Wuhan Province of China to a global pandemic that has affected everyone in the world either directly or indirectly [1]. In the United States alone, approximately 29 million cases and nearly 500,000 deaths have arisen as a result [2].

Agencies, such as the Center for Disease Control (CDC) and the World Health Organization (WHO), have collected data concerning many different demographics of the world's population, which provide a picture of how differently the COVID-19 pandemic and other diseases affect different people, so doctors, first responders, and everyday citizens can make sense of how this disease is spreading and how to combat the rising case numbers and misinformation. The pandemic outbreak

can provide an opportunity to gain critical information for research and developing treatment strategies. Visualizations can be used to generate an understanding of complex data and extract helpful information [3]. Collecting diverse data sets from various sources enables researchers to develop robust visualizations that will allow a more precise story of the data to be told. An article by Anthony Fauci states, "Every outbreak provides an opportunity to gain important information" [4].

Nearly two years after the first lockdown was announced in the United States, vaccines have been developed by a variety of companies [5]. Knowing the pattern and number of the cases that react to how the vaccine is rolled out on a state level can result in critical insights into how state and local governments address this evolving situation. Visualizations can reveal the hidden patterns in the collected data and fill the gaps where there is incomplete or missing data, particularly where marginalized groups or areas are concerned [6]. Studies have shown how detrimental the effects of this disease are to certain demographics; knowing statistical information about a state's population and how the disease spreads could aid the ability to roll out the vaccines and alleviate mandates effectively [7].

Visualizing the data details can reveal hidden information, such as how each state or territory handles its vaccination roll-out and response to an overworked public health system [8]. If more variables are considered, a better understanding can be reached. Viewing the data in a format that provides intuitive feedback will allow more effective local, state, and federal decision-making. However, due to the complicity and dynamic feature, it is still challenging to thoroughly understand such pandemic data and extract critical information, which could delay the speed of providing scientific support for front-line medical workers and government agencies to make the most appropriate and timely decisions [9], [10].

To address the challenges, we seek to explore the coronavirus data by considering multi-dimensional variables. The focus of this paper is to visualize and analyze the data collected from various sources. With the raw information, a web-based platform and algorithms have been developed to represent all data sets in one cohesive visualization. Due to the effectiveness of color and mapping diagrams possessing the ability to convey information hidden in raw data visually [11], the visualization featured in this paper is a dynamic choropleth map to provide an easy way to visualize how

pandemic varies across a geographic area or show the level of variability within a region. At the same time, a multi-source synchronization scheme is developed to link the quantitative charts with a colored map and enhance all the selected graph components simultaneously. A bar chart overlay method is designed to show multi-dimensional variables in one graphic visualization. The connected scatterplot is combined with a bar chart to visually compare data acquired from multiple sources, enabling users to read the numbers for each state while also maintaining the ability to see how the disease trends overall and keep sight of both the forest and the trees as it were. The developed data dynamic visualization platform seeks to improve risk perceptions by introducing interaction as a critical component, which will aid professionals in communicating the importance of these pandemic data while helping regular citizens make safe and conscious decisions in a world with coronavirus disease.

TABLE I

DATA SAMPLE OF THE FIRST 10 STATES (ALPHABETICAL ORDER) ACQUIRED FROM [12]: VACCINATION AT LEAST ONE DOSE, FULLY VACCINATED, BOOSTER, AND TOTAL COVID CASE AND DEATH RATE.

State	Id	One Dose	Fully Vaccinated	Booster	Total Case	Total Death
Alabama	AL	64	52	20	31.2	1.4
Alaska	AK	72	64	31	41.1	0.5
Arizona	AZ	76	64	29	33.3	1.4
Arkansas	AR	69	56	24	32.6	1.3
California	CA	85	74	42	29.8	0.9
Colorado	CO	82	72	40	30.0	0.8
Connecticut	CT	95	82	49	26.2	1.2
Delaware	DE	86	72	35	32.5	1.0
Dis. of Col.	DC	95	82	38	25.2	0.8

TABLE II

DATA SAMPLE OF THE 6 STATES FROM 11 TO 16 (ALPHABETICAL ORDER) ACQUIRED FROM [13]: THE PERCENTAGE OF AGE GROUP 0-18, 19-25, 26-34, 55-64, AS WELL AS 65 AND OVER.

State	Id	0-18	19-25	26-34	35-54	55-64	65 & Over
Florida	FL	20.8	7.7	11.2	25.1	13.9	21.3
Georgia	GA	25.0	8.6	12.2	26.6	12.6	14.9
Hawaii	HA	22.6	7.1	11.5	25.2	13.2	20.4
Idaho	ID	26.1	8.6	11.7	24.8	12.1	16.7
Illinois	IL	23.5	8.5	12.1	26.2	13.2	16.6
Indiana	IN	24.6	8.9	11.9	25.1	13.1	16.4

TABLE III

DATA SAMPLE OF THE 6 STATES FROM 17 TO 22 (ALPHABETICAL ORDER) ACQUIRED FROM [14]: THE PERCENTAGE OF FULLY VACCINATED IN THE AGE GROUP 5-11, 12-17, 18-64, AND 65 AND UP.

State	Id	5-11	12-17	18-64	65 & Up
Iowa	IA	27	51	68	95
Kansas	KS	27	54	70	95
Kentucky	KY	21	45	64	88
Louisiana	LA	14	40	60	87
Maine	ME	46	71	86	95
Maryland	MD	47	80	85	95

TABLE IV

DATA SAMPLE OF THE 6 STATES FROM 47 TO 52 (ALPHABETICAL ORDER) ACQUIRED FROM [14]: THE PERCENTAGE OF UPDATED BOOSTER IN THE AGE GROUP UNDER 18, 18-64, AND 65 AND UP.

State	Id	Under 18	18-64	65 & Up
Vermont	VT	23	52	88
Virginia	VA	12	40	74
Washington	WA	13	44	77
West Virginia	WV	4	26	60
Wisconsin	WI	10	39	81
Wyoming	WY	4	22	61

II. MATERIALS AND METHODS

A. Platform and Tools Used

Visual Studio Code is used for code development. Skypack and Cloudflare are used to load additional JavaScript libraries. The programming languages used include HTML, CSS, and JavaScript. The JavaScript libraries used include D3, d3-scale, and d3-legend. Google Chrome is used as the web browser to load the visualization.

An open-source, cross-platform JavaScript runtime environment Node.js [15] is used to build the server that runs JavaScript and server-side scripting, which enables an asynchronous event-driven JavaScript runtime, and Node.js web application framework Express [16] is employed to provide features for web application and launch client-side data visualization in web browsers.

B. Data Preparation and Processing

The multiple data sets are taken from the New York Times “The Coronavirus Pandemic” [12], the State Health Facts [13], and Centers for Disease Control and Prevention (CDC) [14], respectively. The collected data are stored in four corresponding CSV (comma-separated values) files, which are text files that have a specific format allowing data to be saved in a table structured format.

Table I shows a sample data of the first 10 states (alphabetical order) acquired from [12], including the state names and corresponding Ids, as well as the population percentage of vaccination, i.e., at least one dose, fully vaccinated, updated booster, and the total coronavirus disease rate and death rate due to Covid-19 at the time October 20, 2022. The data of the population distribution by age is scrapped from [13]. The data set contains the population percentage of 5 age groups in each state in the year 2021, and the 6 states from 11 to 16 are illustrated in Table II. The data of Covid-19 vaccinations in the United States by defined age groups are demonstrated in Tables III and IV, which contains 6 selected states to show information of fully vaccinated and updated booster in the age group, respectively.

The *d3.csv(url, callback)* function in D3 is used to send http request for fetching the file of type csv at the specified URL and execute the callback function with parsed csv data objects, which can be used as multidimensional arrays for data processing and calculation. In this project, we save the csv files in Node.js server’s local folder. The *d3.json()* is

employed to send http request to the specified URL to load JSON (JavaScript Object Notation) file of United States that contains 51 states. The state name in the JSON file is applied to match the loaded CSV file for data processing and color mapping in each state. The data preparation and processing are described in Fig. 1.

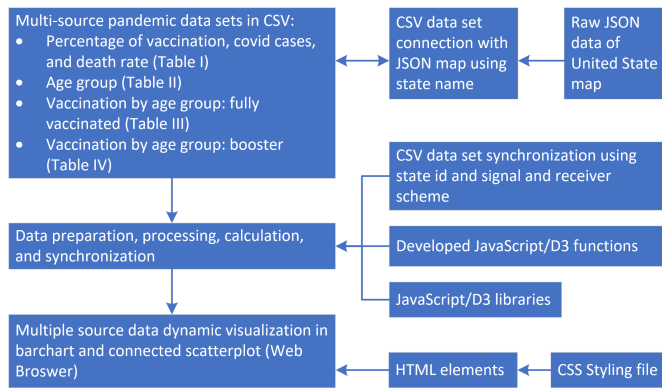


Fig. 1. The flowchart shows the pipeline of synchronizing multiple raw data in CSV format and a JSON map of the United States, data processing, calculation, component transformation, and dynamic visualization in a web browser.

C. Data Visualization and Synchronization

In the developed platform, each graph chart is created on an independent SVG (Scalable Vector Graphics) element, which is rendered with corresponding JavaScript function. As shown in Fig. 2, five graph charts are rendered simultaneously to display the acquired data sets listed in Table I-III, and the data set columns of the percentage of vaccination, Covid-19 cases, death due to coronavirus disease are dynamically mapped to the United States map illustrated in Fig. 2 (a). The mapping is controlled the menu, which is created with HTML select element. The “As least one dose” is the default mapping, and users can interactively change the mapping selection, and the 5 mapping graphs are displayed in Fig. 3. When users change the mapping from the system menu, the SVG element is cleared, and the graph chart is rendered again with the new data.

The JavaScript portion of the code is contained in an HTML script element, and each data set is set a unique id and select by D3, and a SVG (Scalable Vector Graphics) element is appended to the id for 2D graphics rendering. As demonstrated in Fig. 2 (b)-(c), our algorithm displays multiple data columns on one SVG element, which greatly improves the efficiency of space usage and quantitative information comparison. We also combine 2 display methods connected scatterplot and bar chart on one SVG element, which is illustrated in Fig. 2 (d)-(e) and used for information synchronization.

The connected scatterplot and bar chart use the same linear scale, mapped from zero to the maximum value in the current column, as selected by the second HTML select element. The mapped values are used in the y-axis, and the state names are extracted by their ids and display on the x-axis. The data map uses a linear scale of mapping the minimum and maximum

values of the current column to yellow and blue colors, respectively. On this scale, a blue color $rgb(0, 102, 235)$ represents a larger value, while a yellow color $rgb(255, 255, 120)$ represents a smaller value. All the chart graphs are rendered in their individual JavaScript functions.

To create the connected scatterplot and bar chart graphs, the data are used to calculate the width of each rectangle, and the index of each value in the data set is used to calculate the y attribute. The largest values are shown at the top of the bar graph, and the smallest are shown towards the bottom. Both the data map and bar graph represent only one column from one data set at a time. Different instances of the visualization are used to compare different variables. The flowchart of the rendering process is shown in Fig. 1. The initial rendering of the visualization is shown in Fig. 2.

The JavaScript element events *mouseover* and *mouseleave* are used to synchronize the data enhancement among all the display graphs. When mouse is on a display SVG element, which is identified by the state id, all the graph elements corresponding to this id will receive a signal about the selection and their color is set to cyan, which is demonstrated in Fig. 3. When the mouse is moved out of the selected element, a signal is sent out to all these elements about color changes, and all the corresponding elements will recover their original colors. This event-based scheme used in our platform will dynamically synchronize the state enhancement among all the displayed graphs in real time.

III. RESULTS AND ANALYSIS

Multiple data sets and their associated variables were rendered with quantitative numbers to investigate the relationships among vaccination and coronavirus disease cases and the death rate due to Covid-19. The initial visualization web page is illustrated in Fig. 2.

A. Information Enhancement

The software platform introduces a synchronization function to enhance feedback using event and signal scheme. When users move mouse to hover on any bars or state in one of the six graphs, which is corresponding to one state, all the other five graphs will enhance to corresponding state information with cyan color, so the users can easily see and analyze the information in all these graphs. The quantitative information is also displayed in the graphs and charts, so we can visualize and synchronize multi-source data sets in one web browser, and the enhancement can be dynamically updated using mouse.

The Fig. 3 demonstrates this function using the state Texas as an example, from which we can see that state Taxes is colored with cyan in (a), the age group of this state is also enhanced in (b), and the corresponding vaccination and booster information as well as the coronavirus case rate and death rate are also colored in (c) and (d), respectively. At the same time, users can also intuitively detect the enhanced quantitative information of the vaccinations by age group, which is illustrated in (e) and (f) for showing the percentage

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At Least One Dose 2022

The purpose of this visualization is to show the percentage of vaccination in each state of United State, including at least one dose, fully vaccinated, and booster enhancement, as well as the covid case rate and death rate due to covid in each state. The percentage of each age is also displayed. The connections of these factors are extrated and analyzed.

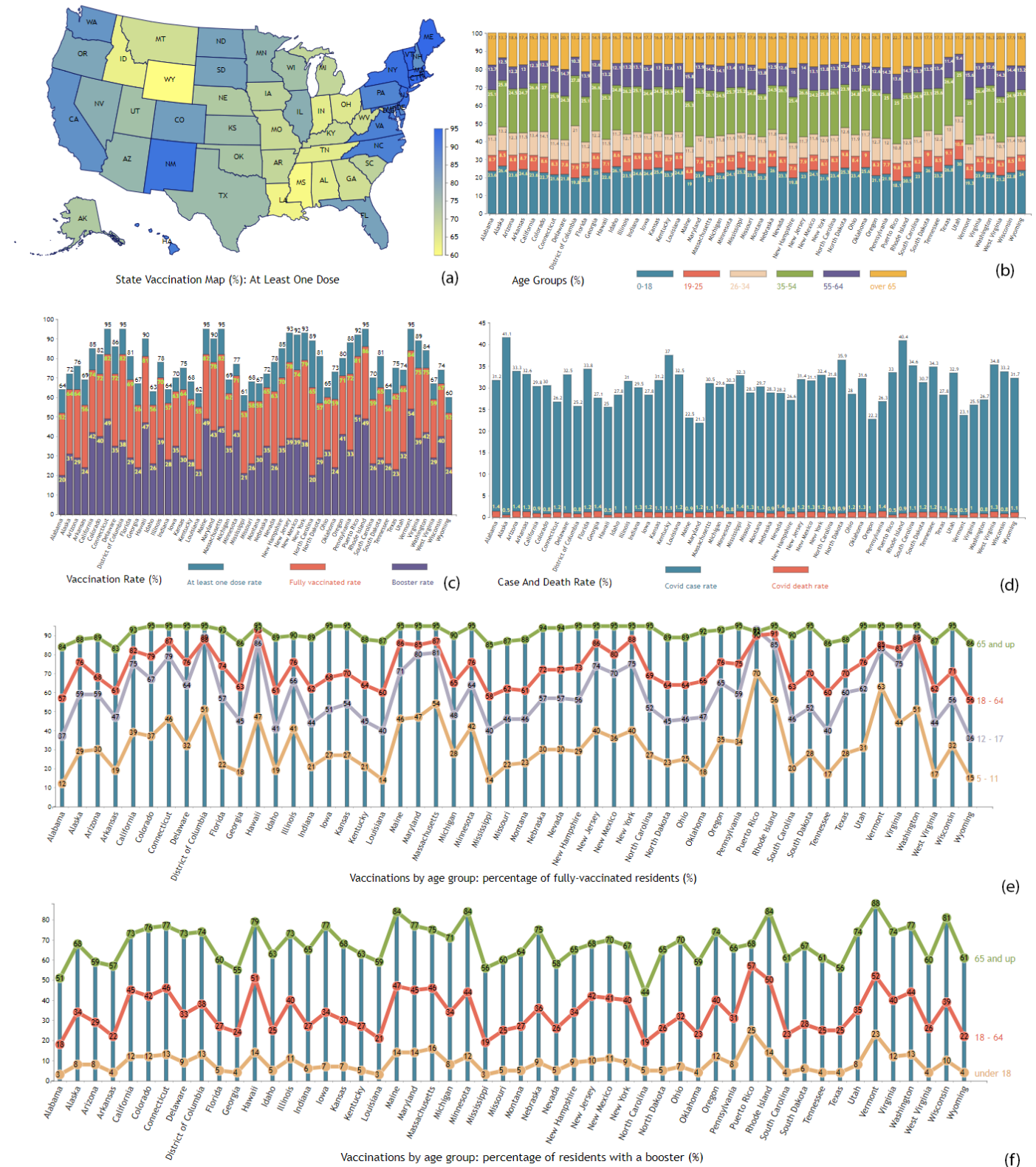


Fig. 2. The initial rendering of the visualization platform of the main user interface in a web browser, including the United States map with each state enhanced with different scaled colors, the state-based age group population, vaccination population, and booster percentage, coronavirus disease case and death rate, and the vaccination rate distributions among various age groups.

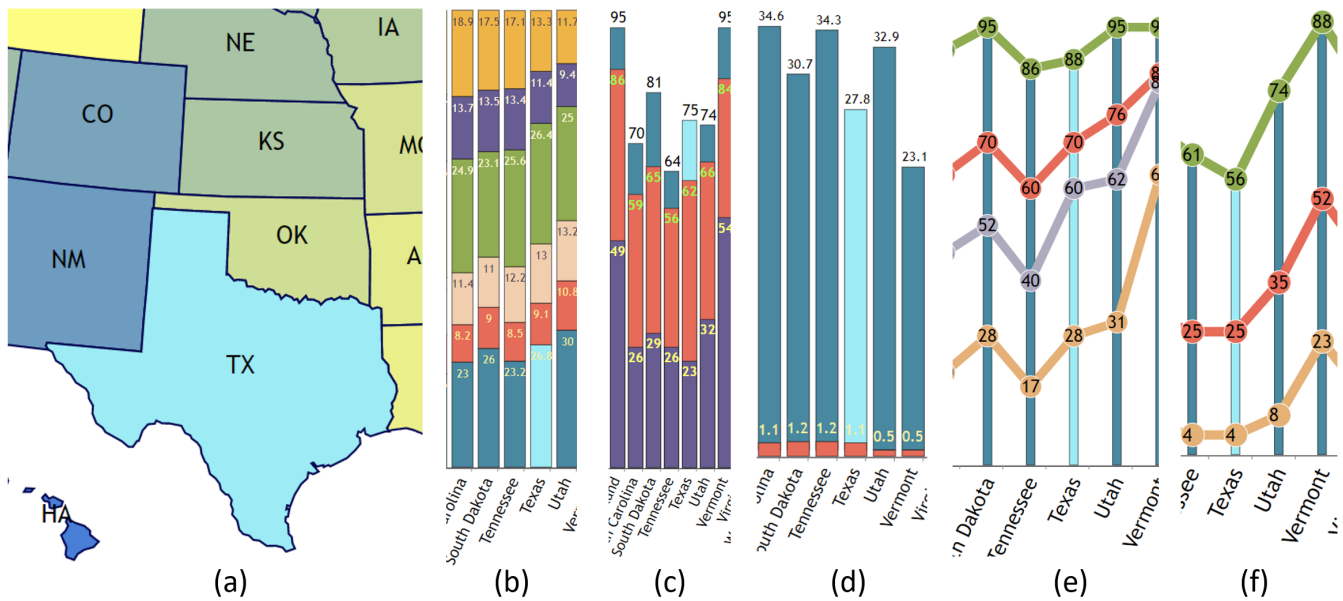


Fig. 3. Demonstration of multi-source synchronized visualization using Texas state as an example, which is dynamically enhanced with cyan color, including (a) state vaccination map, (b) age group, (c) vaccination rate (at least one dose, fully vaccinated, and booster), (d) covid case and death rate, (e) vaccination percentage by age group of fully vaccinated, as well as (f) vaccination percentage by age group of booster enhancement.

of full-vaccinated residents and percentage of residents with a booster, respectively.

B. Data Mapping

The quantitative information visualized in Fig. 2(b), (c), and (d) can be mapped to the United States map (a) using color mapping, including the categories of “At Least One Dose”, “Fully Vaccinated”, “Booster”, “Case Rate”, and “Death Rate”, and the mapping result is illustrated in Fig. 4(a)-(e), respectively. The mapping domain is the minimum and maximum for the data column in the corresponding date set, and the mapped range is from yellow to blue, so the ranges of the scale bar on the right in each of the five maps are different, for example, the range is from 60 to 95 in the at least one dose graph, while in the covid death rate graph, the range is from 0.5 to 1.4. Therefore, we can easily detect the color difference in the states in the map.

From Fig. 4(a)-(d) we can see that when a state has higher at least one dose rate, this state is usually has higher fully vaccinated and booster rate, and vice versa. When compare (c)-(d) with (d) and (e), we can detect that when a state has higher vaccination rate, this state usually has relatively lower case rate and corresponding lower death rate due to coronavirus disease. Moreover, we can see that the death rate due to Covid-19 is low, which is less than 1.4%. We can conclude that Covid-19 has relatively strong transmission capability, i.e., can be around 40% in the total population, but the mortality risk is low.

We also get information that the vaccination is region based, for example, the northeast region has relatively higher vaccination and booster rate, which can achieve around 90%~95%, and this region also has relatively lower coronavirus disease

case rate that is less than 25%. On the other hand, the south region, especially, Louisiana, Mississippi, Alabama, Tennessee, Kentucky, and West Virginia, has relatively lower vaccination rate of around 60% fully vaccinated, and the region has relatively higher case rate that is around 32%~37%.

C. Multi-Source Bar Chart

The quantitative information of vaccination rate including at least one dose, fully vaccinated rate, and booster can be displayed in one bar chart in our system, which is described in Fig. 2(c). We use merge method to design this visualization, during which the largest data set, i.e., at least one dose, is display first in the bar chart in dark cyan color, then the second largest data set, i.e., fully vaccinated, is displayed in the same bar chart in light red color, so in each state, the light red color is covered to the dark cyan color, next, we display the smallest data set, i.e., booster rate, in blue color. The quantitative numbers in each state of the three categories are shown in each bar, so we can analyze and compare these information between the states.

The similar method is used to visualize the quantitative information of case rate and death rate in Fig. 2(d). We can detect the relationship of vaccination rate and case rate using these two bar charts. For example, in the states Maine, Maryland, and Vermont, the fully vaccinated rates are 82%, 78%, and 84%, respectively, and the corresponding case rates are 22.5%, 21.3%, and 23.1%, respectively. On the contrary, in the state Tennessee, Arkansas, and Louisiana, the fully vaccinated rates are 56%, 52%, and 55%, respectively, and the corresponding case rates are 34.3%, 31.2%, and 32.5%, respectively. There are some exceptional cases, such as the states Rhode Island and Massachusetts, the fully vaccinated

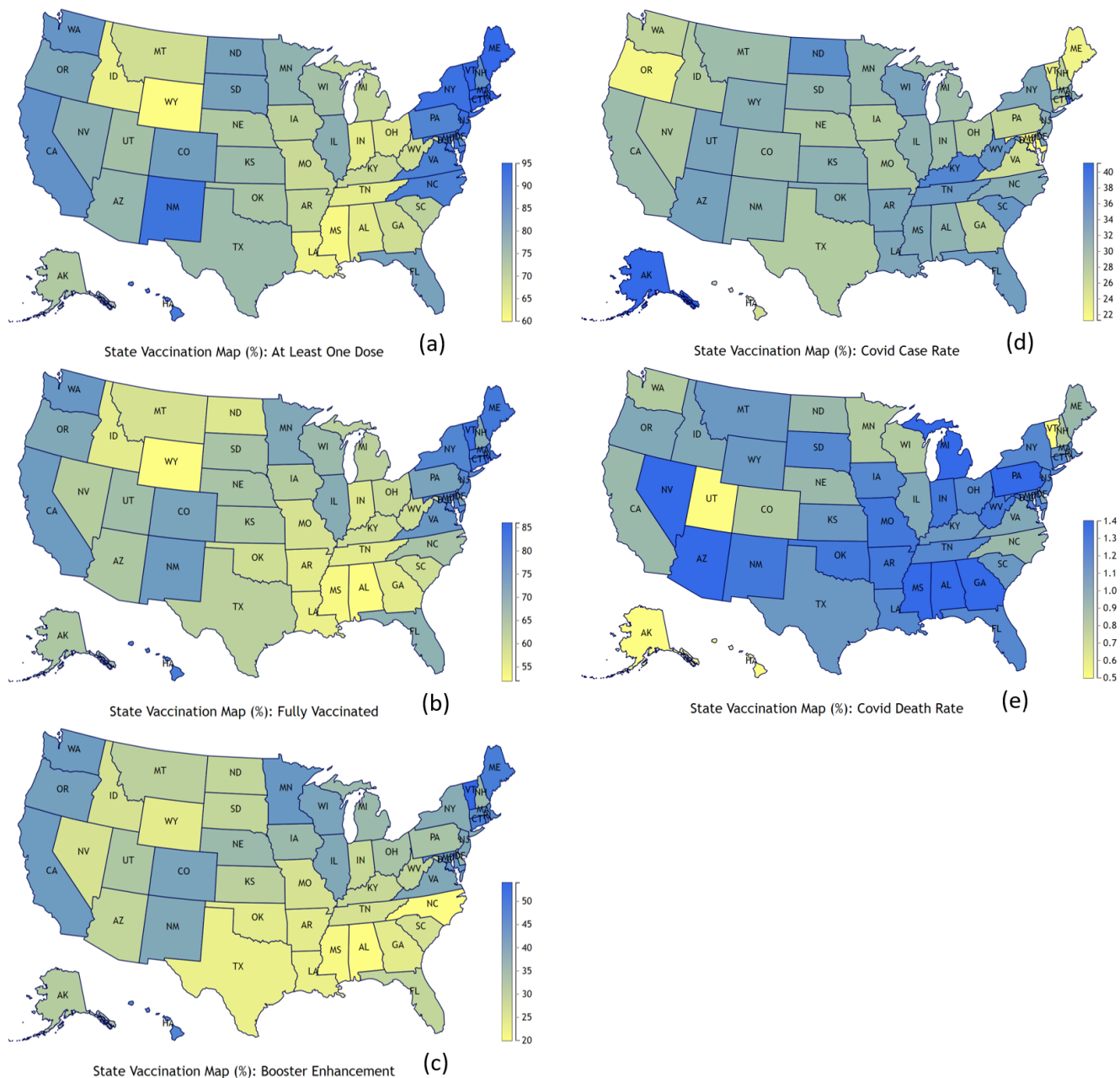


Fig. 4. The United States map representation of the percentage of vaccination, coronavirus case rate, and death rate due to Covid-19: (a) at least one dose, (b) fully vaccinated, (c) updated booster, (d) coronavirus case rate, and (e) death rate.

rates are 86% and 78%, but the corresponding case rates are 40.4% and 30.5%, so we also need to consider other factors that can affect the case rate.

D. Age Group Consideration

Usually the coronavirus disease rate has connections with the percentage of age groups in the population of a state, so we show the age group of all the 52 states in the United States. The percentage of six age groups displayed in Figs. 2(b), i.e., 0-18, 19-25, 26-34, 35-54, 55-64, and 65 and over. However, we did not detect the direct relationship between the groups

in a state with its coronavirus disease rate. Therefore, we also display the vaccination and booster rates in the age groups of all the 52 states, which are illustrated in Figs. 2(e) and (f), respectively.

The connected scatterplot is used with bar chart to display the vaccination and booster rate in various age groups. The bar chart is used to connect different age groups in a state and also to synchronize information enhancement, i.e., when mouse is moved over the bar of state, the state's visual information in all other graphs are enhanced simultaneously in cyan color,

which has been demonstrated in Fig. 3 using Texas as an example. As demonstrated by Fig. 2(e), for the percentage of fully vaccinated residents, four age groups visualized, i.e., 5-11, 12-17, 18-64, and 65 and up, while for the percentage of residents with a booster category, three age groups are displayed, i.e., under 18, 18-64, and 65 and up, which is illustrated in Fig. 2(f). From both (e) and (f), we can see that for both fully-vaccinated and booster rate, the age group 65 and up has significantly higher rate than the other age groups, which can achieve above 84% and 51% in all states, while the corresponding rates for the age group 18-64 are 67% and 18%, respectively. The age group of under 18 is much lower than the first two age groups.

From the vaccination rate in the age group analysis we can conclude that even the senior residents have relatively higher chances to get coronavirus disease, but this age group has much higher vaccination and booster rate, so Covid-19 case rate is not higher than other age groups. We can also know that the reason of no direct connections between the percentage of age group and the state's coronavirus disease rate.

IV. CONCLUSION

This research aims to develop new and dynamic visualizations that consider coronavirus disease and vaccination data in various dimensions. To address the issue of lacking published technologies in connecting and synchronizing multiple sets of pandemic data in one web page for comparison and analysis, we developed algorithms and a web-based software platform for dynamically displaying pandemic data acquired from multiple sources and visually extracting information such as vaccination, coronavirus disease rate, percentage of mortality, the vaccination distributions among various age groups, and their relationships with virus transmission rate.

Through manipulating the SVG (Scalable Vector Graphics) elements with D3 as an extension of JavaScript, we can visualize and map the multi-source data with various categories of coronavirus disease, vaccination, and booster interactively. Because of the use of JavaScript functions and HTML select elements, the choropleth map, bar chart, and connected scatterplot can be dynamically rendered on one web page, allowing multi-dimensional variables to be examined without needing to change the code and reload the visualization. Once the graphs and charts are generated, various collective instances of the visualization can be compared to search for patterns and extract hidden information in the pandemic data.

A complete picture of the data can be drawn by comparing the number of fully vaccinated citizens to the total reported coronavirus cases and death rate scaled by the state's population. Considering more than one variable, such as how both the vaccine rollout and patients act as a function of time, will allow for more robust analyses by statisticians and healthcare professionals. This paper is not meant to explain why the data behaves the way it does; this research developed new algorithms to show the data in a more accessible environment. Knowing this information could help more effective decisions

as the pandemic continues and the world returns to normalcy while keeping safety as a number one concern.

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